



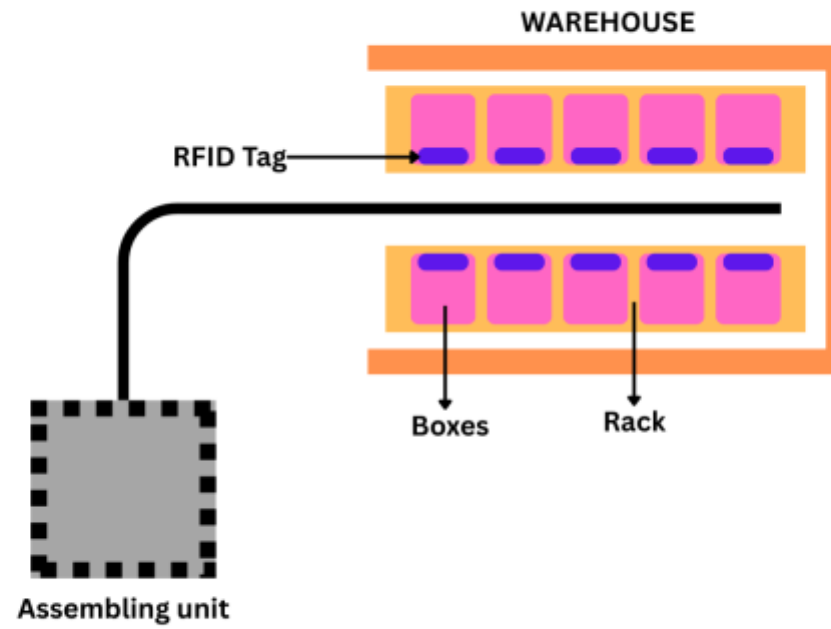
Warehouse Automation: Line Follower Robot

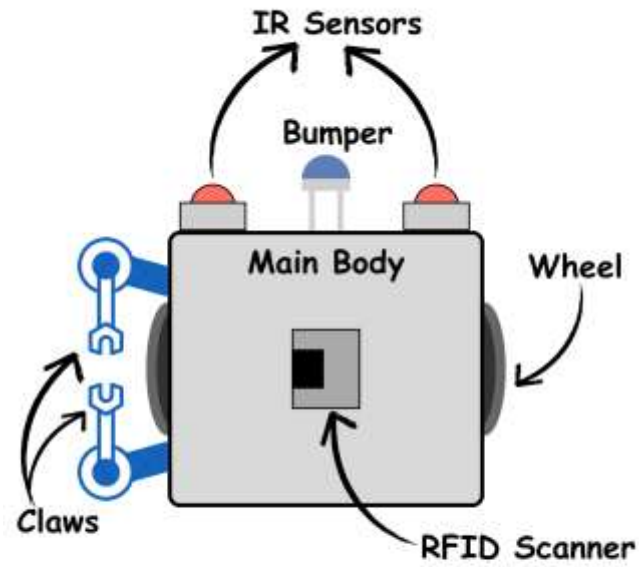
Guide: Prof. Mamta Khosla



The problem:

An autonomous robot is assigned an RFID (which maps to a box of cargo) to look for in the warehouse. Each box is identified by a unique RFID. The bot is to manoeuvre itself to the warehouse, find the correct box, and bring it back to the assembly unit.







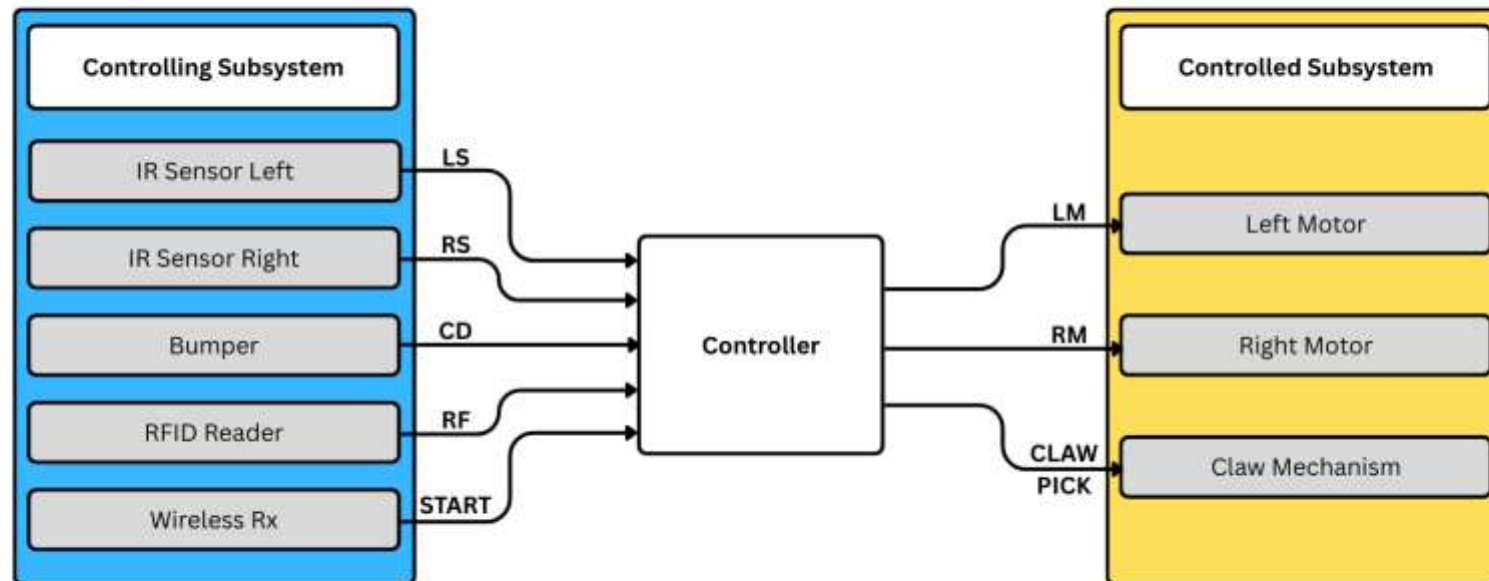
Our Proposed Solution Workflow:

- High Level Block Diagram
- Flowchart
- Mnemonic Documented State Diagram
- Code assignment, Minimization using K-maps and Circuit implementation
- Implementation of controller using Verilog

1) Block Diagram



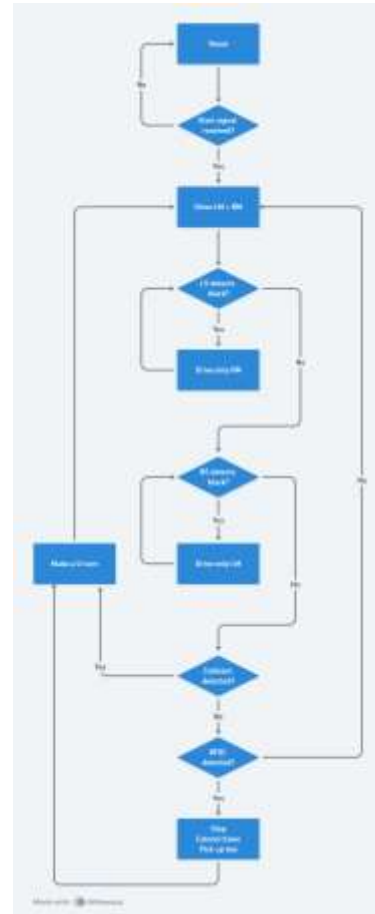
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2) Flowchart



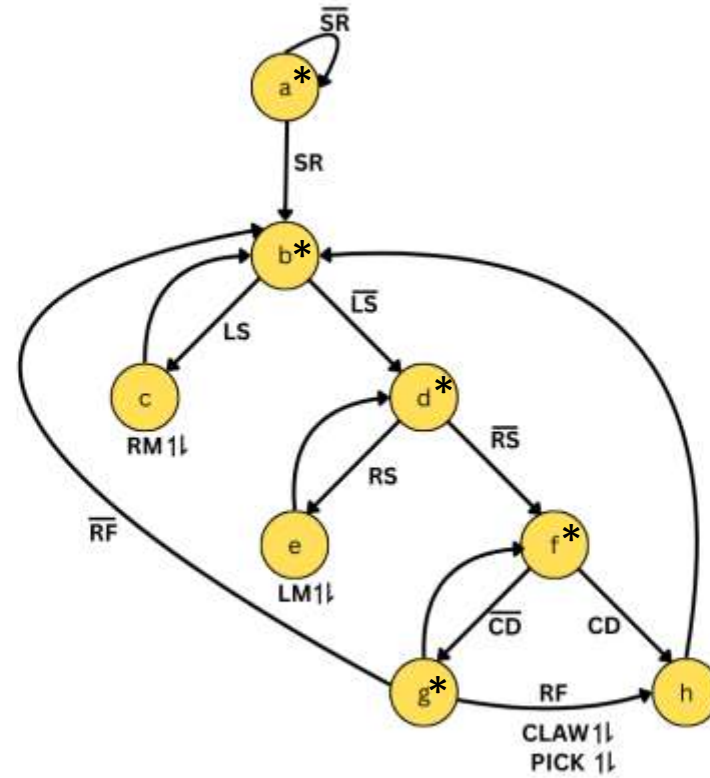
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3) Mnemonic Documented State Diagram



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4.1) Code Assignment



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Code Assignment K-Map:

A \ BC				
	00	01	11	10
0	a	b	h	g
1	c	d	e	f

Adjacency Table:

Previous State	Present State	Next State
a*	a*	(a, b)*
a*, c, g*, h	b*	(c, d)*
b*	c	b
b*, e	d*	(e, f)*
d*	e	d
d*	f*	(g, h)*
f*	g*	(b, h)*
(f, g)*	h	b*

4.2) Minimization using K-map



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C \ AB	00	01	11	10
	0	0	0	RF
0	0	0	0	RF
1	0	1	0	1



C \ AB	00	01	11	10
	0	0	0	0
0	0	0	0	0
1	0	1	0	1

$$D_A = A'B.RF + A'BC + A'BC$$

C \ AB	00	01	11	10
	SR	(LS)'	1	1
0	SR	(LS)'	1	1
1	1	RS	1	CD



C \ AB	00	01	11	10
	0	0	1	1
0	0	0	1	1
1	1	0	1	0

$$D_B = A'B'.SR + BC'.LS + BC.RS + AB'.CD + AC' + AB + A'B'C$$

4.2) Minimization using K-map



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AB		00	01	11	10
C	0	0	1	0	0
	1	0	1	1	0

$$D_C = A'B + BC$$

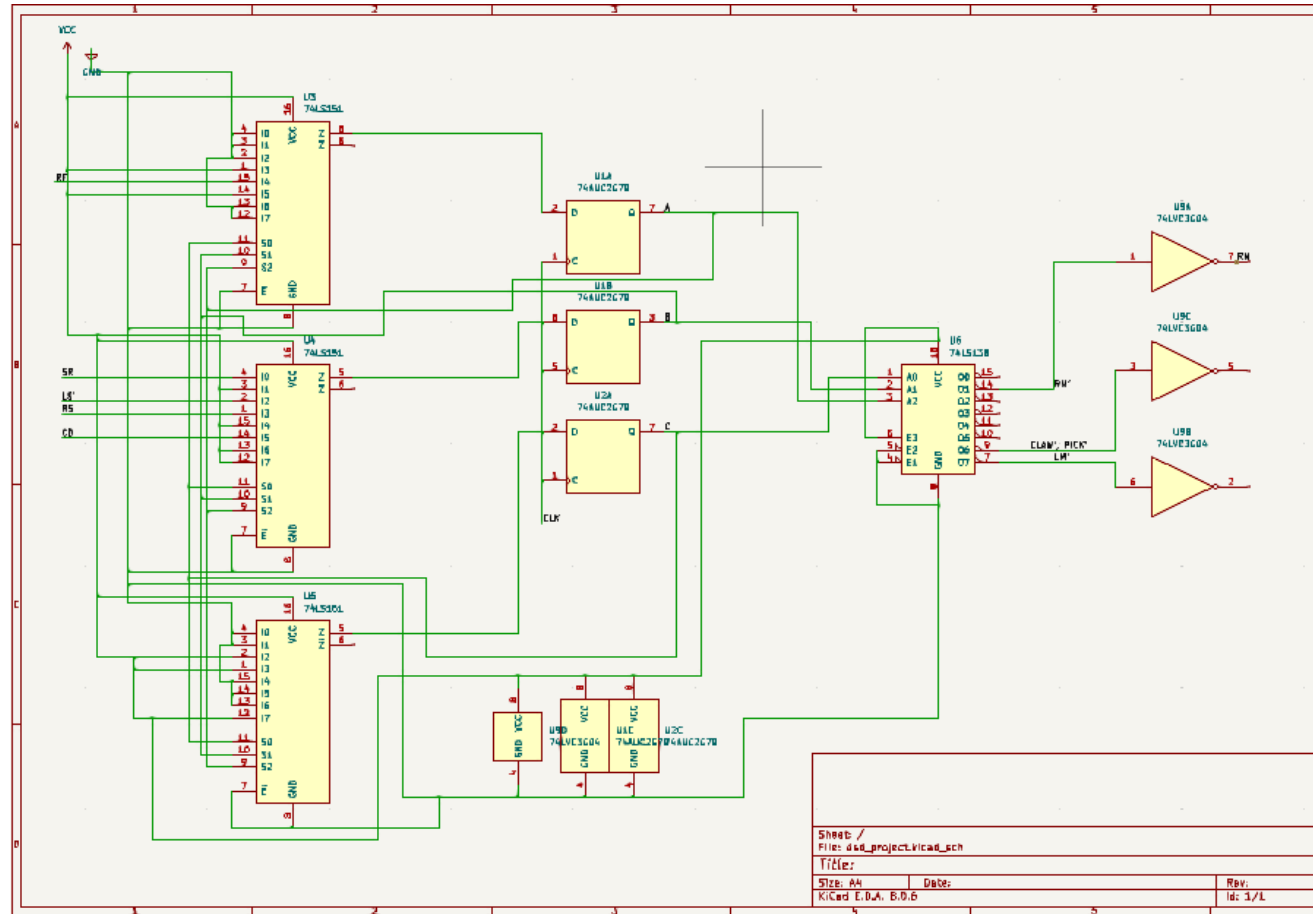
Output expressions:

$$RM = A'B'C$$

$$LM = ABC$$

$$CLAW = ABC'.RF$$

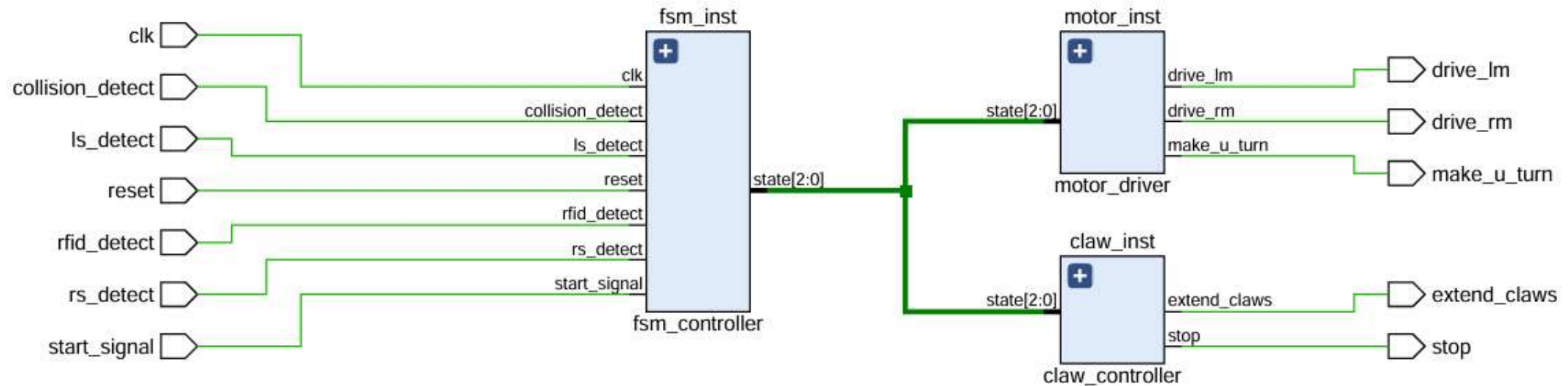
$$PICK = ABC'.RF$$



5.1) Circuit Implementation (Top level view)



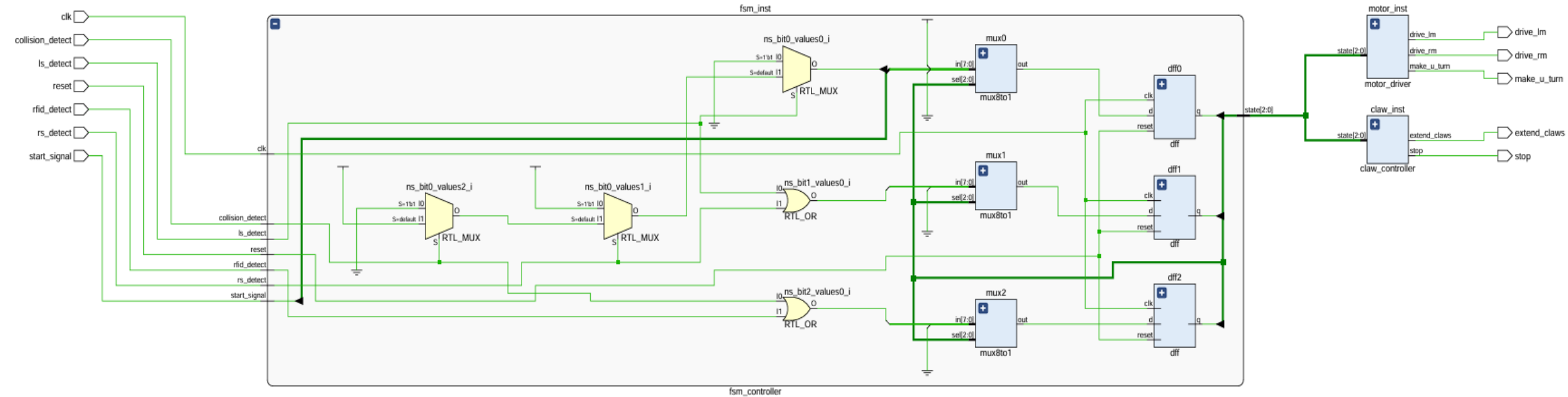
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5.2) Circuit Implementation (In Depth schematic)



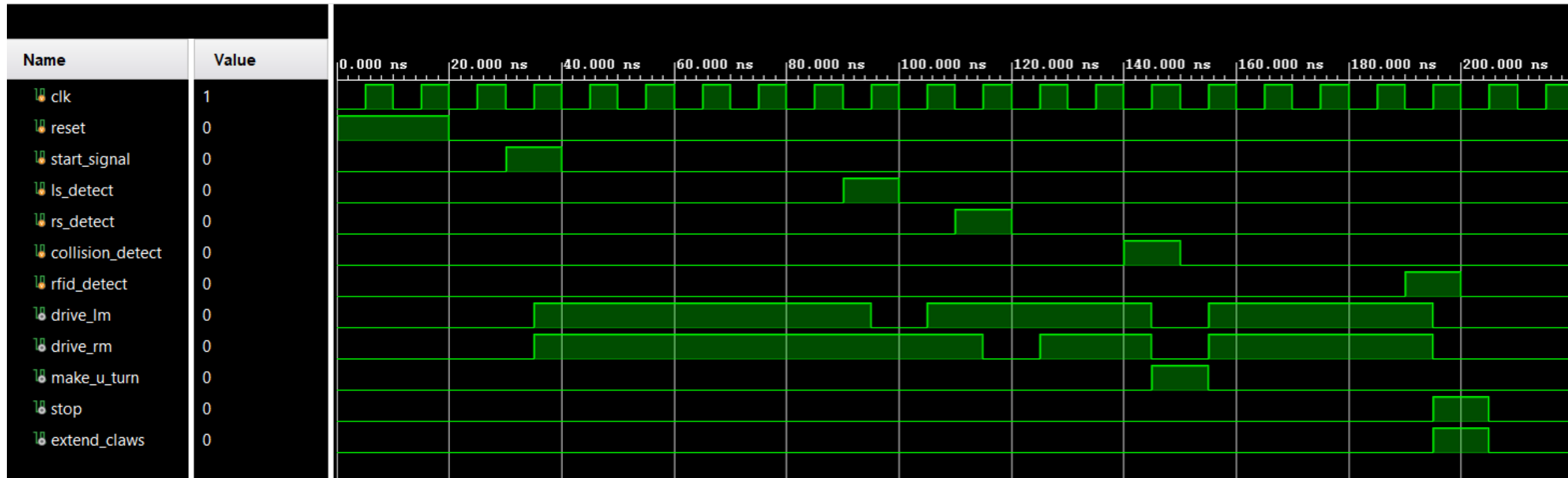
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5.3) Circuit Implementation (Timing Diagram)



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Predefined Modules:

D-Flip Flop

```
// dff.v
module dff (
    input wire clk,
    input wire reset,
    input wire d,
    output reg q
);
    always @(posedge clk or posedge reset) begin
        if (reset)
            q <= 1'b0;
        else
            q <= d;
        end
    endmodule
```

8:1 MUX

```
// mux8to1.v
module mux8to1 (
    input wire [7:0] in,      // 8 input lines
    input wire [2:0] sel,     // 3-bit selector
    output wire out           // Output
);
    assign out = in[sel]; // Simple and synthesizable
endmodule
```



Functional Modules:

Claw controller

```
module claw_controller (  
    input wire [2:0] state,  
    output reg stop,  
    output reg extend_claws  
);  
  
always @(*) begin  
    stop = 0;  
    extend_claws = 0;  
  
    if (state == 3'b101) begin // PICK_BOX  
        stop = 1;  
        extend_claws = 1;  
    end  
end  
endmodule
```

Motor Driver

```
module motor_driver (  
    input wire [2:0] state,  
    output reg drive_lm,  
    output reg drive_rm,  
    output reg make_u_turn  
);  
  
always @(*) begin  
    drive_lm = 0;  
    drive_rm = 0;  
    make_u_turn = 0;  
  
    case (state)  
        3'b001: begin // DRIVE_BOTH  
            drive_lm = 1;  
            drive_rm = 1;  
        end  
        3'b010: drive_rm = 1; // DRIVE_RM_ONLY  
        3'b011: drive_lm = 1; // DRIVE_LM_ONLY  
        3'b100: make_u_turn = 1; // U_TURN  
        default: ; // No motor action for other states  
    endcase  
end  
endmodule
```

6) Notes and References



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- Source Code: <https://github.com/Chinmay-Deo-2004/Warehouse-Automation-Bot-Controller>
- Theory: *An Engineering Approach to Digital Design (William L. Fletcher)*
- Software used:
 - Xilinx Vivado
 - KiCad